

# PROJECT TEST PLAN

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## Online Quiz System

Team 73  
PES University  
September 2026

### 1. Document Information

| Item         | Details                                                  |
|--------------|----------------------------------------------------------|
| Project      | Online Quiz System                                       |
| Team No.     | 73                                                       |
| Team Members | Nakshathira; Poojitha P.; Sharanya N.; Rahul Rajkumar S. |
| Document     | Project Test Plan                                        |
| Basis        | Software Requirements Specification (SRS), Version 1.0   |
| Prepared For | Software Engineering Project Work                        |

### 2. Test Plan Overview

The Online Quiz System is a web-based application for conducting multiple-choice quizzes. The system allows users to select quiz categories, attempt timed quizzes, receive scores and feedback, and view leaderboard performance. Administrators manage quiz questions and categories.

This test plan defines the testing approach for verifying the functional and non-functional requirements specified in the SRS. Testing will verify that quiz operations, scoring, result storage, leaderboard functions, administration, authentication, security, usability, reliability, and responsive behavior work as specified.

### 3. Test Objectives

- Verify that all functional requirements REQ-01 to REQ-19 are implemented correctly.
- Verify performance requirements REQ-20 and REQ-21 under normal operation.
- Verify authentication, authorization, and secure transmission requirements REQ-22 to REQ-24.
- Verify that quiz results and submitted answers are stored accurately.
- Verify that expired quizzes do not accept additional responses.
- Verify usability, reliability, maintainability, testability, and responsive behavior.
- Identify defects before release and record their status through the project defect process.

### 4. Test Scope

#### 4.1 In Scope

- User registration/login and authentication.
- Quiz category selection and quiz selection.
- Timed multiple-choice quiz execution.
- Answer selection and recording.
- Automatic quiz termination on timeout.
- Scoring and instant feedback.

- Result storage and result display.
- Leaderboard generation and ranking.
- Administrator question and category management.
- Performance, security, usability, reliability, and browser/device responsiveness.

## 4.2 Out of Scope

The SRS does not specify third-party payment systems, social login, mobile-native applications, or external analytics. These are therefore outside this test plan unless added to the requirements later.

## 5. Test Strategy

| Testing Level / Type  | Purpose                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------|
| Unit Testing          | Verify individual modules such as authentication, quiz timer, scoring, and question management. |
| Integration Testing   | Verify interactions among UI, application server, and database.                                 |
| System Testing        | Verify the complete Online Quiz System against the SRS.                                         |
| Functional Testing    | Verify each functional requirement and expected user/admin behavior.                            |
| Performance Testing   | Check response time and accurate recording during normal operation.                             |
| Security Testing      | Verify authentication, authorization, credential protection, and HTTPS behavior.                |
| Usability Testing     | Verify clear navigation, readable content, consistent controls, and useful messages.            |
| Compatibility Testing | Verify operation on commonly used modern desktop and mobile web browsers.                       |
| Regression Testing    | Re-test affected functions after defect fixes or requirement changes.                           |
| Acceptance Testing    | Confirm that the implemented system satisfies the project requirements.                         |

## 6. Test Environment

| Component   | Test Environment                                                                                |
|-------------|-------------------------------------------------------------------------------------------------|
| Client      | Desktop and mobile web browsers                                                                 |
| Browser     | Modern web browsers                                                                             |
| Application | Web-based Online Quiz System                                                                    |
| Server      | Application server hosting quiz logic and result processing                                     |
| Database    | Structured database for users, categories, quizzes, questions, answers and results              |
| Network     | Internet/network connection using HTTP/HTTPS; HTTPS preferred for sensitive data                |
| Test Data   | Sample users, administrator account, categories, quizzes, questions, answer options and results |

## 7. Functional Test Cases

| ID    | Test Area        | Req.   | Test Condition          | Expected Result                                                     | Priority |
|-------|------------------|--------|-------------------------|---------------------------------------------------------------------|----------|
| TC-01 | Category display | REQ-01 | User opens quiz system  | Available quiz categories are displayed.                            | High     |
| TC-02 | Select category  | REQ-02 | User selects a category | Selected category is accepted and associated quizzes are displayed. | High     |
| TC-03 | Display quizzes  | REQ-03 | Category selected       | Quizzes associated with the selected category are displayed.        | High     |
| TC-04 | Display MCQs     | REQ-04 | User starts a quiz      | Multiple-choice questions for the selected quiz are displayed.      | High     |
| TC-05 | Answer options   | REQ-05 | Question displayed      | Multiple answer options are shown for each question.                | High     |

|       |                             |        |                                              |                                                                          |        |
|-------|-----------------------------|--------|----------------------------------------------|--------------------------------------------------------------------------|--------|
| TC-06 | Start timer                 | REQ-06 | User starts quiz                             | Timer starts when quiz begins.                                           | High   |
| TC-07 | Record answers              | REQ-07 | User selects answers                         | Selected answers are recorded correctly.                                 | High   |
| TC-08 | Auto-end timeout            | REQ-08 | Timer reaches limit                          | Quiz automatically ends and no further responses are accepted.           | High   |
| TC-09 | Evaluate answers            | REQ-09 | Quiz submitted                               | Submitted answers are compared with correct answers.                     | High   |
| TC-10 | Instant feedback            | REQ-10 | Answer submitted/completion occurs           | Appropriate feedback is provided.                                        | High   |
| TC-11 | Calculate score             | REQ-11 | Answers evaluated                            | Score is calculated from the number of correct answers.                  | High   |
| TC-12 | Display final score         | REQ-12 | Quiz completed                               | Final score is displayed.                                                | High   |
| TC-13 | Store score                 | REQ-13 | Quiz completed                               | User's quiz score is stored.                                             | Medium |
| TC-14 | Leaderboard records         | REQ-14 | Completed results exist                      | Leaderboard records are maintained for completed quizzes.                | Medium |
| TC-15 | Display rankings            | REQ-15 | User opens leaderboard                       | Rankings are displayed based on quiz performance.                        | Medium |
| TC-16 | Add question                | REQ-16 | Admin submits valid new question             | Question is validated and stored.                                        | Medium |
| TC-17 | Update question             | REQ-17 | Admin edits existing question                | Updated question data is stored and reflected.                           | Medium |
| TC-18 | Delete question             | REQ-18 | Admin deletes question                       | Question is removed from available quiz content.                         | Medium |
| TC-19 | Manage categories           | REQ-19 | Admin adds/updates/removes category          | Category changes are validated and stored.                               | Medium |
| TC-20 | Normal response performance | REQ-20 | User performs normal quiz actions            | Questions/actions are provided with minimal delay.                       | Medium |
| TC-21 | Accurate result recording   | REQ-21 | User submits quiz                            | Responses and scores are recorded accurately without noticeable delay.   | High   |
| TC-22 | Authentication              | REQ-22 | Unauthenticated user attempts account access | Authentication is required before accessing user account functions.      | High   |
| TC-23 | Admin authorization         | REQ-23 | Normal user attempts admin function          | Administrative functions are restricted to authorized administrators.    | High   |
| TC-24 | Secure transmission         | REQ-24 | User logs in/transmits sensitive data        | Credentials and sensitive information are protected during transmission. | High   |

## 8. Detailed Boundary and Negative Tests

| ID    | Scenario           | Input/Condition                         | Expected Result                                                         |
|-------|--------------------|-----------------------------------------|-------------------------------------------------------------------------|
| TC-25 | Timer boundary     | Timer reaches exactly 00:00             | Quiz ends at the specified limit; extra answers are rejected.           |
| TC-26 | Invalid login      | Incorrect credentials submitted         | System does not authenticate the user and shows an appropriate message. |
| TC-27 | Unauthorized admin | Normal user accesses admin URL/function | Access is denied.                                                       |

|       |                       |                                                      |                                                                                                |
|-------|-----------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------|
| TC-28 | Empty answer          | User attempts submission without selecting an option | System handles the condition according to defined quiz behavior without corrupting the result. |
| TC-29 | Invalid question data | Admin submits incomplete/invalid question data       | System validates input and does not store invalid content.                                     |
| TC-30 | Repeated submission   | User attempts to submit a completed quiz again       | System prevents duplicate/incorrect result processing.                                         |

## 9. Non-Functional Test Checklist

| Area               | Reference    | Verification                                                                                           |
|--------------------|--------------|--------------------------------------------------------------------------------------------------------|
| Performance        | REQ-20/21    | Normal quiz operations respond with minimal/no noticeable delay; results are recorded accurately.      |
| Safety/Reliability | SRS §6.2     | Submitted answers/results are not lost during normal operation; expired sessions reject new responses. |
| Security           | REQ-22/23/24 | Authentication is enforced, admin functions are restricted, and sensitive transmission is protected.   |
| Usability          | SRS §6.4     | Interface is clear, simple, readable, and navigation/control placement is consistent.                  |
| Maintainability    | SRS §6.4     | System components remain modular enough to support maintenance.                                        |
| Testability        | SRS §6.4     | Requirements and system behaviors are clearly defined and verifiable.                                  |
| Responsiveness     | SRS §6.4     | System provides consistent experience on commonly used desktop and mobile browsers.                    |
| Compatibility      | SRS §7.2     | System works through commonly used modern web browsers.                                                |
| Data consistency   | SRS §7.3     | Stored user and quiz data remain accurate and consistent.                                              |

## 10. Requirement Traceability

| Requirement Group  | Requirements     | Test Cases                     |
|--------------------|------------------|--------------------------------|
| Category Selection | REQ-01 to REQ-03 | TC-01 to TC-03                 |
| Timed Quiz         | REQ-04 to REQ-08 | TC-04 to TC-08, TC-25          |
| Scoring & Feedback | REQ-09 to REQ-12 | TC-09 to TC-12                 |
| Leaderboard        | REQ-13 to REQ-15 | TC-13 to TC-15, TC-30          |
| Content Management | REQ-16 to REQ-19 | TC-16 to TC-19, TC-29          |
| Performance        | REQ-20 to REQ-21 | TC-20 to TC-21                 |
| Security           | REQ-22 to REQ-24 | TC-22 to TC-24, TC-26 to TC-27 |

## 11. Entry and Exit Criteria

### 11.1 Entry Criteria

- Testable build of the Online Quiz System is available.
- Required database and test data are available.
- Functional requirements and expected behaviors are identified.
- Test environment is accessible.

## 11.2 Exit Criteria

- All planned test cases have been executed.
- All critical/high-severity defects are resolved or formally accepted by the team.
- Required functional requirements have corresponding test evidence.
- No known unresolved defect prevents normal quiz operation.
- Test results are documented for submission/review.

## 12. Defect Reporting

| Field                 | Description                                             |
|-----------------------|---------------------------------------------------------|
| Defect ID             | Unique identifier for the defect.                       |
| Title                 | Short description of the issue.                         |
| Requirement/Test Case | Related requirement and test case ID.                   |
| Steps to Reproduce    | Steps required to reproduce the defect.                 |
| Expected Result       | Behavior specified by the requirement.                  |
| Actual Result         | Observed system behavior.                               |
| Severity              | Critical / High / Medium / Low.                         |
| Priority              | Order in which the team plans to fix the defect.        |
| Status                | New / In Progress / Fixed / Retest / Closed / Reopened. |
| Evidence              | Screenshot, log, or other supporting evidence.          |

## 13. Sample Test Execution Record

| Test Case ID | Actual Result                   | Status  | Defect ID |
|--------------|---------------------------------|---------|-----------|
| TC-01        | To be recorded during execution | Not Run | —         |
| TC-06        | To be recorded during execution | Not Run | —         |
| TC-08        | To be recorded during execution | Not Run | —         |
| TC-11        | To be recorded during execution | Not Run | —         |
| TC-15        | To be recorded during execution | Not Run | —         |
| TC-23        | To be recorded during execution | Not Run | —         |
| TC-24        | To be recorded during execution | Not Run | —         |

## 14. Test Data

The following controlled test data will be used to execute the test cases. These are sample values for testing and do not represent production user data.

| Data Category   | Sample Test Data                                   | Purpose                                           |
|-----------------|----------------------------------------------------|---------------------------------------------------|
| User Account    | Username: testuser01; valid test password          | Normal user authentication and quiz participation |
| Invalid User    | Username: invaliduser; incorrect password          | Negative authentication testing                   |
| Administrator   | Username: admin01; valid administrator credentials | Administrative authorization testing              |
| Quiz Categories | Java; DBMS; Python                                 | Category selection and category management        |
| Quiz            | Java Basics Quiz                                   | Quiz selection and execution                      |

|                       |                                                       |                                       |
|-----------------------|-------------------------------------------------------|---------------------------------------|
| Questions             | 5 sample MCQs with four options each                  | Question display and answer selection |
| Correct Answers       | One defined correct option for each test question     | Scoring and answer evaluation         |
| Incorrect Answers     | Incorrect options for selected questions              | Feedback and score verification       |
| Timer                 | Normal test: 10 minutes; boundary test: 1 minute      | Timer and timeout testing             |
| Leaderboard Data      | Multiple completed quiz results with different scores | Ranking verification                  |
| Invalid Question Data | Missing question text/option/correct answer           | Input validation testing              |

## 15. Pass/Fail Criteria

Each test case shall be marked according to the comparison between the expected and actual result.

| Result  | Criteria                                                                                                                          |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|
| PASS    | The actual result completely matches the expected result and the requirement is satisfied.                                        |
| FAIL    | The actual result differs from the expected result, the requirement is not satisfied, or a defect prevents the expected behavior. |
| BLOCKED | The test cannot be executed because a required environment, build, dependency, account, or test data is unavailable.              |
| NOT RUN | The test has not yet been executed.                                                                                               |
| RETEST  | The test previously failed and is being executed again after a defect fix.                                                        |

Overall testing may be considered complete when all planned tests have been executed, critical/high-severity defects are resolved or formally accepted, and no known defect prevents normal quiz operation.

## 16. Test Execution and Pass/Fail Record

The following record is used during actual execution. 'Actual Result' and 'Pass/Fail' must be updated by the team after testing; they are intentionally not pre-filled as PASS because the system has not yet been executed.

| ID    | Test Area        | Req.   | Expected Result                     | Actual Result  | Pass/Fail | Defect ID |
|-------|------------------|--------|-------------------------------------|----------------|-----------|-----------|
| TC-01 | Category display | REQ-01 | Available categories displayed      | To be recorded | Not Run   | —         |
| TC-02 | Select category  | REQ-02 | Category accepted                   | To be recorded | Not Run   | —         |
| TC-03 | Display quizzes  | REQ-03 | Associated quizzes displayed        | To be recorded | Not Run   | —         |
| TC-04 | Display MCQs     | REQ-04 | MCQs displayed                      | To be recorded | Not Run   | —         |
| TC-05 | Answer options   | REQ-05 | Options displayed for each question | To be recorded | Not Run   | —         |
| TC-06 | Start timer      | REQ-06 | Timer starts with quiz              | To be recorded | Not Run   | —         |
| TC-07 | Record answers   | REQ-07 | Selected answers saved              | To be recorded | Not Run   | —         |
| TC-08 | Auto-end timeout | REQ-08 | Quiz ends at time limit             | To be recorded | Not Run   | —         |
| TC-09 | Evaluate answers | REQ-09 | Answers evaluated correctly         | To be recorded | Not Run   | —         |

|       |                       |                         |                                      |                |         |   |
|-------|-----------------------|-------------------------|--------------------------------------|----------------|---------|---|
| TC-10 | Instant feedback      | REQ-10                  | Feedback provided                    | To be recorded | Not Run | — |
| TC-11 | Calculate score       | REQ-11                  | Score calculated correctly           | To be recorded | Not Run | — |
| TC-12 | Display final score   | REQ-12                  | Final score displayed                | To be recorded | Not Run | — |
| TC-13 | Store score           | REQ-13                  | Score stored                         | To be recorded | Not Run | — |
| TC-14 | Leaderboard records   | REQ-14                  | Records maintained                   | To be recorded | Not Run | — |
| TC-15 | Display rankings      | REQ-15                  | Rankings displayed correctly         | To be recorded | Not Run | — |
| TC-16 | Add question          | REQ-16                  | Question added and stored            | To be recorded | Not Run | — |
| TC-17 | Update question       | REQ-17                  | Question updated                     | To be recorded | Not Run | — |
| TC-18 | Delete question       | REQ-18                  | Question deleted                     | To be recorded | Not Run | — |
| TC-19 | Manage categories     | REQ-19                  | Category changes stored              | To be recorded | Not Run | — |
| TC-20 | Performance           | REQ-20                  | Minimal delay under normal operation | To be recorded | Not Run | — |
| TC-21 | Result recording      | REQ-21                  | Responses/scores recorded accurately | To be recorded | Not Run | — |
| TC-22 | Authentication        | REQ-22                  | Authentication required              | To be recorded | Not Run | — |
| TC-23 | Admin authorization   | REQ-23                  | Admin functions restricted           | To be recorded | Not Run | — |
| TC-24 | Secure transmission   | REQ-24                  | Sensitive data protected             | To be recorded | Not Run | — |
| TC-25 | Timer boundary        | REQ-08                  | No response accepted after timeout   | To be recorded | Not Run | — |
| TC-26 | Invalid login         | REQ-22                  | Invalid credentials rejected         | To be recorded | Not Run | — |
| TC-27 | Unauthorized admin    | REQ-23                  | Normal user denied admin access      | To be recorded | Not Run | — |
| TC-28 | Empty answer          | Related quiz behavior   | System handles empty answer safely   | To be recorded | Not Run | — |
| TC-29 | Invalid question data | REQ-16                  | Invalid content rejected             | To be recorded | Not Run | — |
| TC-30 | Repeated submission   | Related result behavior | Duplicate processing prevented       | To be recorded | Not Run | — |

## 17. Updated Requirement Traceability Matrix

This matrix provides bidirectional traceability from the SRS requirements to the planned test cases.

| Requirement | Requirement Description               | Mapped Test Case(s) |
|-------------|---------------------------------------|---------------------|
| REQ-01      | Display available quiz categories     | TC-01               |
| REQ-02      | Allow user to select quiz category    | TC-02               |
| REQ-03      | Display quizzes for selected category | TC-03               |
| REQ-04      | Display multiple-choice questions     | TC-04               |
| REQ-05      | Provide multiple answer options       | TC-05               |

|        |                                                   |              |
|--------|---------------------------------------------------|--------------|
| REQ-06 | Start timer when quiz begins                      | TC-06        |
| REQ-07 | Record selected answers                           | TC-07        |
| REQ-08 | Automatically end quiz on timeout                 | TC-08, TC-25 |
| REQ-09 | Evaluate submitted answers                        | TC-09        |
| REQ-10 | Provide instant feedback                          | TC-10        |
| REQ-11 | Calculate score from correct answers              | TC-11        |
| REQ-12 | Display final score                               | TC-12        |
| REQ-13 | Store user's quiz score                           | TC-13        |
| REQ-14 | Maintain leaderboard records                      | TC-14        |
| REQ-15 | Display rankings                                  | TC-15        |
| REQ-16 | Allow admin to add questions                      | TC-16, TC-29 |
| REQ-17 | Allow admin to update questions                   | TC-17        |
| REQ-18 | Allow admin to delete questions                   | TC-18        |
| REQ-19 | Allow admin to manage categories                  | TC-19        |
| REQ-20 | Minimal delay during normal operation             | TC-20        |
| REQ-21 | Accurate response/score recording                 | TC-21        |
| REQ-22 | Require authentication                            | TC-22, TC-26 |
| REQ-23 | Restrict admin functions                          | TC-23, TC-27 |
| REQ-24 | Protect sensitive information during transmission | TC-24        |

## 18. Final Submission Checklist

- ☐ Project Test Plan file reviewed by all four team members.
- ☐ Test cases are mapped to SRS requirements.
- ☐ Test data is prepared before execution.
- ☐ Actual results and Pass/Fail status are filled during execution.
- ☐ Security, performance, usability, reliability, and compatibility tests are considered.
- ☐ Defects found during execution are recorded with evidence.
- ☐ Final file name follows the faculty format: Team 73\_Project Test Plan.